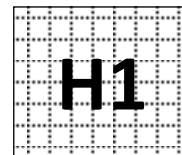


Civics Group	Index Number	Name (use BLOCK LETTERS)
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ST. ANDREW'S JUNIOR COLLEGE
2013 Preliminary Examination

H1 BIOLOGY

8875/2

Paper 2: Core (Mark Scheme)

Thursday

29 August 2013

2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Compulsory question to be answered on writing paper provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/11
2	/8
3	/8
4	/13
5/6	/20
Total	/60

This document consists of **13** printed pages.

[Turn over

Section A

Answer **all** questions.

QUESTION 1

(a) Fig. 1.1 is an electron micrograph showing part of a pancreatic cell.

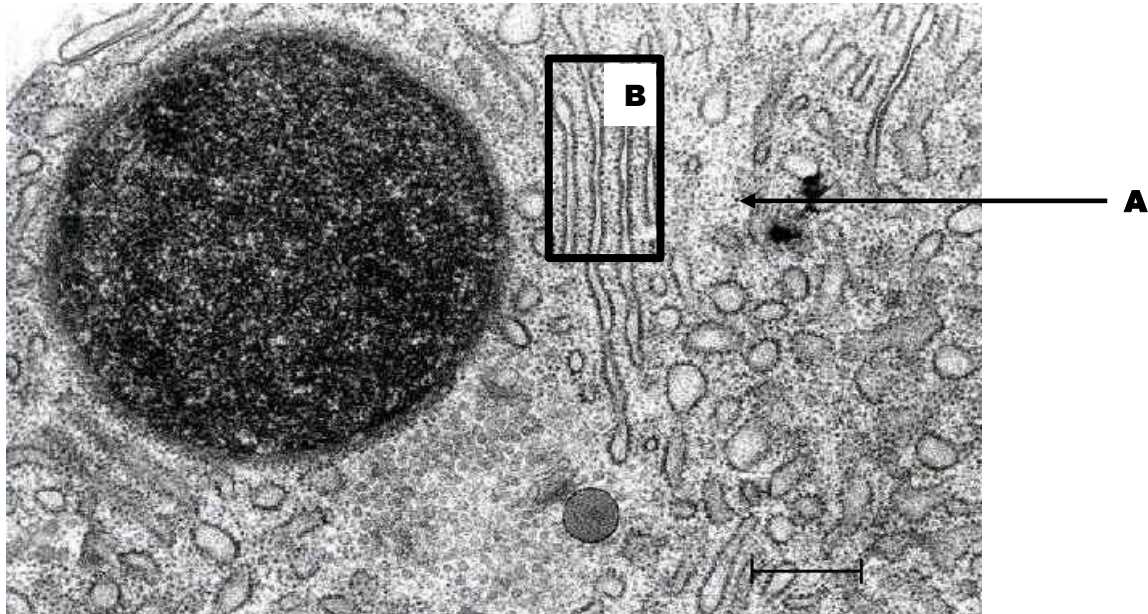


Fig. 1.1

(a)(i) Identify organelle A and describe its specific roles in the production of protein.

.....[2]

- 1 Free ribosomes;
- 2 hold tRNA and mRNA in close proximity;
- 3 position new amino acid to growing polypeptide chain;
- 4 catalyse formation of peptide bond;
- 5 ref. synthesis of proteins that will **function in the cytosol**;

(ii) Certain mitochondria inhibitors can inhibit the functioning of stalked particles. Explain the effects of such inhibitors on organelle B.

.....[1]

- 1 No synthesis of ATP by mitochondria stalked particles;
- 2 Required for protein synthesis/translation at the ribosomes on surface of rER / post-translational modifications (in organelle **B**/rER);

(b) (i) Certain proteins were specifically imported into organelles such as chloroplasts. Suggest and explain how the membrane structure of chloroplast facilitates this process.

.....[2]

- 1 Proteins cannot cross the hydrophobic core of the phospholipid bilayer of mitochondrial membrane;
- 2 Presence of hydrophilic protein transporters/carriers/channels (embedded in membrane);
- 3 Presence of receptor / recognition proteins;
- 4 Complementary in shape to the respective proteins to be imported;

(ii) An aqueous solution contains small membranous vesicles which may be derived from a thylakoid membrane. In these vesicles, the orientation of the ATP synthase enzymes is preserved as it is in the cell.

Fig. 1.2 shows the various experimental systems that are prepared in an attempt to synthesize ATP.

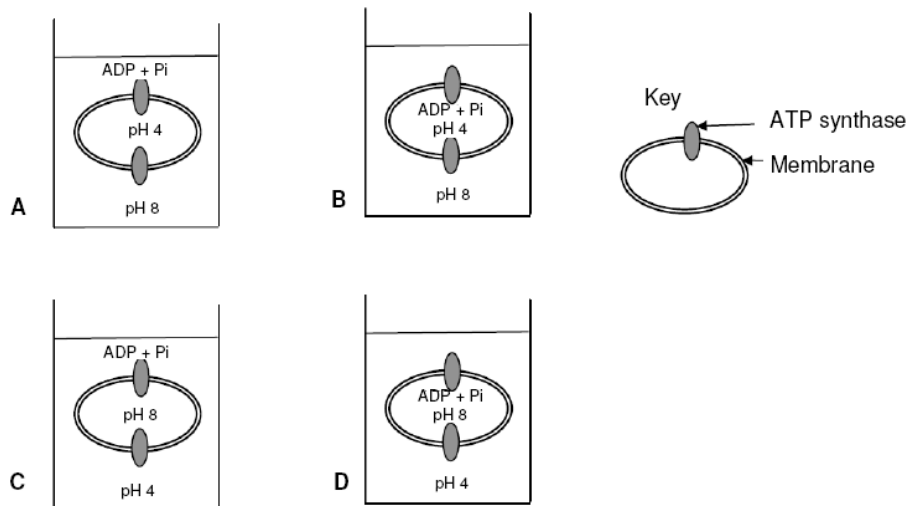


Fig. 1.2

Which experimental system (A to D) shown in Fig. 1.2 would result in the successful synthesis of ATP by thylakoid-derived vesicles? Explain your answer.

.....[3]

- 1 Experimental system A
- 2 pH within the membrane is pH 4 as compared to pH 8 outside the membrane
- 3 Concentration of H^+ ions is higher within the membrane than outside the membrane
- 4 ref. proton gradient
- 5 tendency for the H^+ ions to leave the interior of the vesicle down their concentration gradient
- 6 via ATP synthase;
- 7 ATP is formed using ADP and P_i found outside the membrane

(c) Figure 1.3 shows the effects of increasing substrate concentration on the rate of enzyme reaction for an enzyme functioning in the chloroplast.

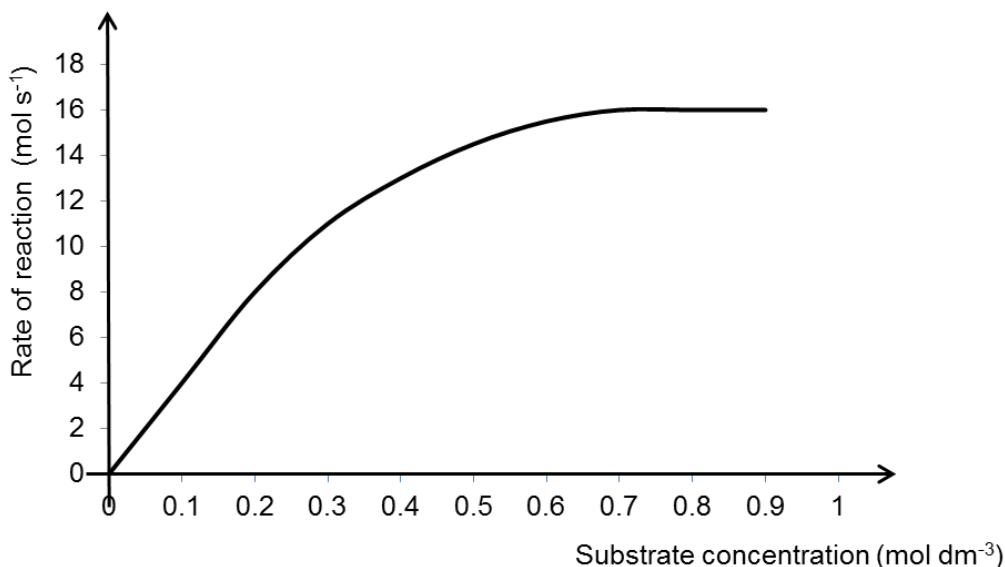


Fig. 1.3

(i) With reference to Fig 1.3, **account** for the **rate of reaction** from **0 to 0.3 mol dm⁻³** of substrate concentration.

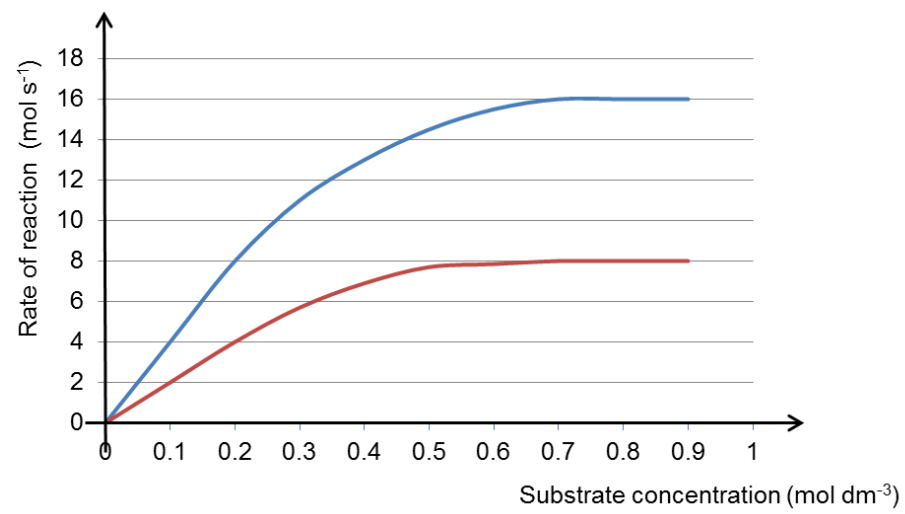
.....[2]

- 1 substrate conc. increases, rate of reaction increases;
- 2 from 0 to 10 mol s⁻¹ (from 0 to 0.3 mol dm⁻³ of substrate concentration);
- 3 increase in **frequency of effective/successful collisions** between enzyme and substrate molecules;
- 4 more enzyme-substrate complexes formed;
- 5 ref. enzyme is in excess / substrate is limiting;

(ii) Exposure of the enzyme to chemical compound X affects the rate of reaction negatively. Analyses revealed that compound X has a **shape complementary to a domain away from the active site**, and has a low level of structural similarity between actual substrate and compound X. **Draw** on Fig. 1.3, the prediction of the rate of reaction of the enzyme after subjection to chemical compound X.

.....[1]

- 1 K_m of 0.2 mol dm⁻³ which is the same as that of Graph without inhibitor;
- 2 V_{max} cannot be attained;



[11 marks]

QUESTION 2

Bt crops are named after *Bacillus thuringiensis* (Bt), a bacteria that naturally produces a crystal protein that is toxic to many pest insects. Bt crops are genetically engineered to produce the Bt toxin in every cell of the plant, with the goal of protecting the crops from pests.

To kill pests, ingested Bt toxins must first become activated by gut protease, before binding to a toxin-binding cadherin protein in the midgut. This cadherin protein is normally localized in the membrane and functions as a receptor for the toxin.

Mutation of the protease gene and/or the cadherin gene in the pests can result in a phenotype resistant towards Bt crops. Both genes are inherited in a recessive manner.

(a) Explain how these mutations result in resistance to Bt crops.

.....[2]

- 1 different primary structure / sequence of amino acids coded for after the mutation would be different
- 2 protein with different tertiary structure / 3D conformation
- 3 structure of active site of protease is changed (thus non-functional protease cannot activate the toxin in the midgut)
- 4 structure of BT binding site of cadherin receptor is changed (thus non-functional cadherin protein cannot localize in the membrane / cannot bind to toxin)
- 5 ref. to BT no longer complementary to protease and cadherin receptor;
- 6 insects with mutated alleles are at selective advantage;
- 7 survive, reproduce and pass down the resistant alleles to offspring;

(b) Define the following terms:

(i) genotype

.....[1]

1. the genetic makeup of an organism;
2. the paired alleles that produces a phenotype;
3. can be either homozygous or heterozygous;

(ii) phenotype

.....[1]

1. physical and chemical characteristic of an organism;
2. depend on genotype of the organism and the environment;

A cross between a *Bt* resistant insect (with mutant receptor, inactive protease) and a *Bt* susceptible insect will have a 75% chance of getting a *Bt* resistant offspring. Assume the insects are diploid.

(c) Using the symbols provided below, draw a genetic diagram to show these results.

.....[3]

Symbols:

Allele A: dominant allele for active protease

Allele a: recessive allele for inactive protease

Allele B: dominant allele for normal cadherin protein

Allele b: recessive allele for abnormal cadherin protein

Parental phenotypes:

Parental genotype:

Parental gametes:

Resistant insect

aabb

(ab)

x

Susceptible insect

AaBb

(AB)

(Ab)

(aB)

(ab)

F₁ genotypes (Punnett Square)

Gametes	(ab)
(AB)	AaBb Susceptible
(Ab)	Aabb Resistant
(aB)	aaBb Resistant
(ab)	aabb Resistant

F₁ Phenotypic ratio: Resistant insect : Susceptible insect

3

:

1

1 Parental genotypes ;

2 Parental gametes;

3 F₁ genotypes ;

4 Corresponding F₁ genotype to F₁ phenotype ;

5 F₁ phenotypic ratio ;

6 correct presentation of genetic diagram (circled gametes, headings)

(d) State one **social/ethical concern**, other than pest resistance, pertaining to the use of **Bt crops** in agriculture.

.....[1]

1 GM crops that kill pests could also cause unintentional harm to other organisms;

2 Eg. Pollen from Bt corn could harm larvae from the Monarch butterfly;

OR

3 Concern about safety to consumers;

4 Antibiotic resistant genes used in vectors may be transferred to bacteria in the consumers;

[8 marks]

QUESTION 3

Fig. 3.1 shows a diploid germ cell with 3 pairs of homologous chromosomes during metaphase.

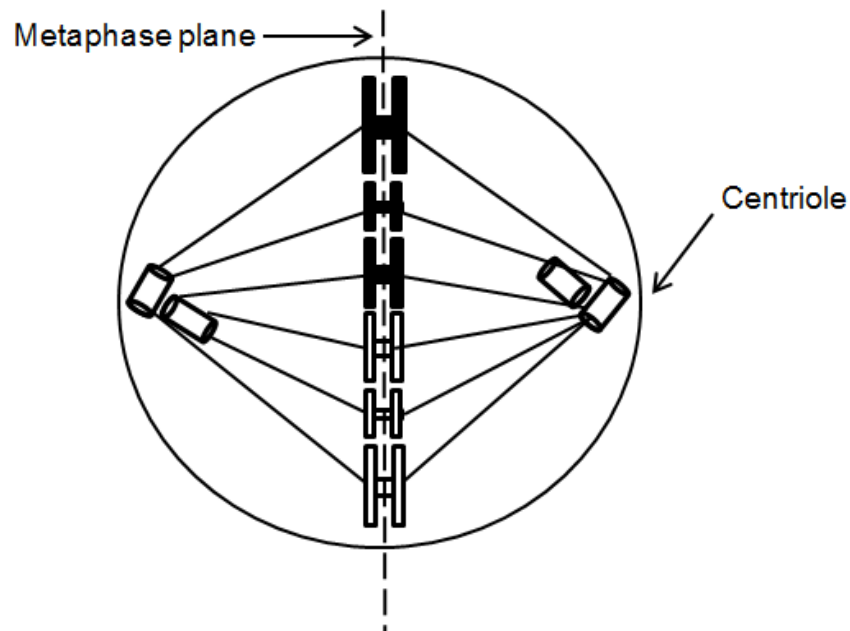


Fig. 3.1

(a) Explain why the homologous chromosomes in Fig. 3.1 do not form bivalents.

.....[1]

- 1 chromosomes are undergoing mitosis;
- 2 no pairing of homologous chromosomes required /
no reduction in the chromosome number required;

(b) With reference to an appropriate property of stem cells, explain if stem cells undergo meiosis.

.....[2]

- 1 Stem cells do not undergo meiosis;
- 2 [Property] **self-renewal** of stem cells;
- 3 requires daughter cells which are **genetically identical** to parental cell;
- 4 with **same number of chromosomes**;
- 5 and **DNA sequence** as the parental cell;

(c) An error occurred **once** during gamete formation in a diploid germ cell of a woman. As a result, her children will have 50% chance of having trisomy of chromosome 9; and 50% are monosomy (only one copy is present) for chromosome 9 in his/her cells. Explain how this could have happened.

.....[2]

1. Non-disjunction;
2. homologous chromosomes fail to separate in anaphase I;

3. so two copies of chromosome 9 will be found in 2 out of 4 oocytes / eggs/ gametes;
4. remaining 2 oocytes/eggs/gametes will have no chromosome 9;
5. upon fertilisation by fusion with the male gamete; (there is 50% chance of having 3 copies of chromosome 9 and 50% chance of having none in the offspring);

Examiner's comments:

Reject: non-disjunction at anaphase II/separation of sister chromatids as an error that occurred once at meiosis II will result in 50% chance of normal (2 copies of chromosome 9); 25% chance of getting 3 copies of chromosome 9; 25% chance of getting only 1 copy of chromosome 9 after fusion with sperm

Fig 3.2 shows a midsection of a DNA molecule during the S phase of interphase.

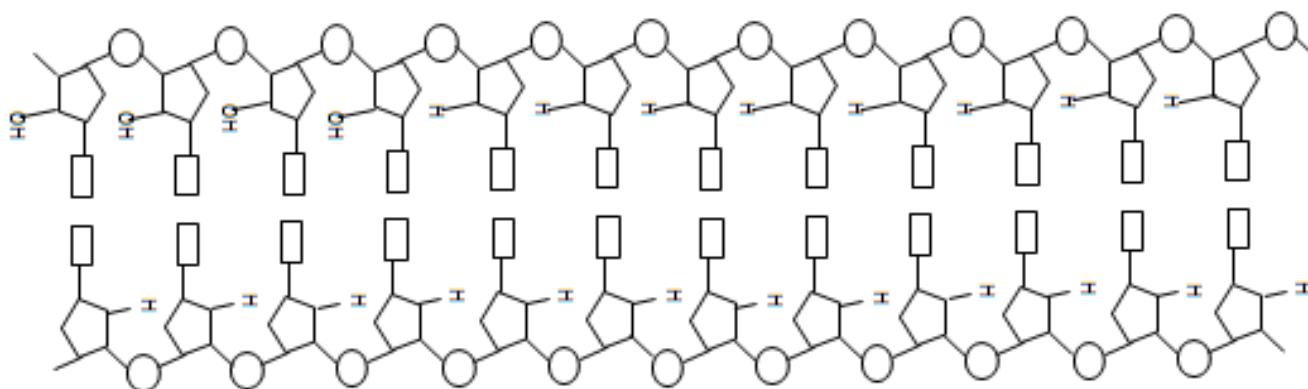


Fig 3.2

(d)(i) Account for the difference in the nucleotides in the top strand compared to bottom strand.

-[2]
- 1 [Difference] Presence of ribonucleotides and deoxyribonucleotides in the top strand while bottom strand has only deoxyribonucleotides;
 - 2 [Explanations] semi-conservative replication is taking place;
 - 3 Bottom strand is used as the template;
 - 4 RNA primer complementary to the template;
 - 5 is synthesized by primase;
 - 6 ref. DNA polymerase III can only add deoxyribonucleotides to the free 3' OH end of an already existing polynucleotide;

(ii) Outline the next immediate step in the further processing of nucleic acid strands shown in Fig 3.2.

-[1]
- 1 DNA polymerase I removes the RNA primer;
 - 2 replacement with **complementary deoxyribonucleotides**;

[8 marks]

QUESTION 4

A student researcher attempted to clone the Human Growth Hormone (HGH) gene. Two separate samples of HGH DNA were prepared and subjected to agarose gel electrophoresis. Results were shown in Fig.4.1.

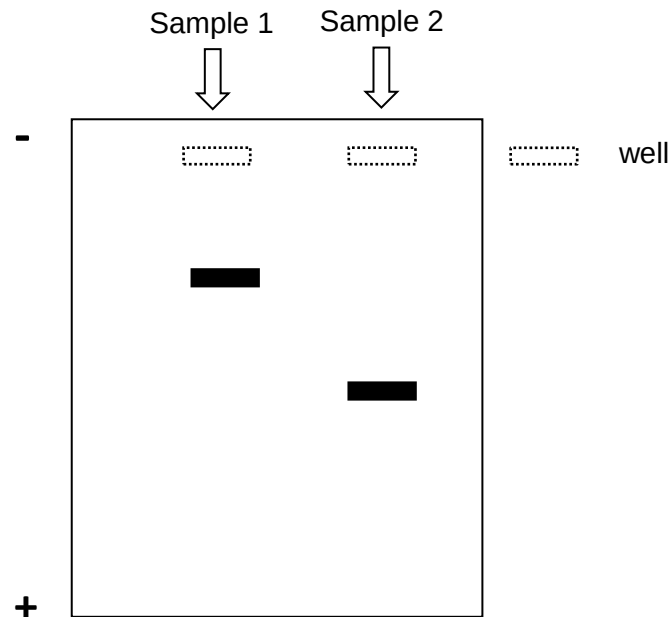


Fig. 4.1

Subsequent cloning procedures involving *Escherichia coli* host cells yield functional HGH protein when Sample 2 was used, but not for Sample 1.

(a)Based on the difference in the bands observed in Fig. 4.1, account for the failure to produce functional HGH protein for Sample 1.

.....[2]

1. sample 1 has a higher molecular weight/larger size compared to sample 2;
2. due to presence of introns in sample 1;
3. bacterial host cells do not have RNA splicing machinery;
4. ref. HGH gene not expressed correctly;

The student researcher has designed the following plasmid (Fig. 4.2) for use in her preliminary trials of a cloning experiment.

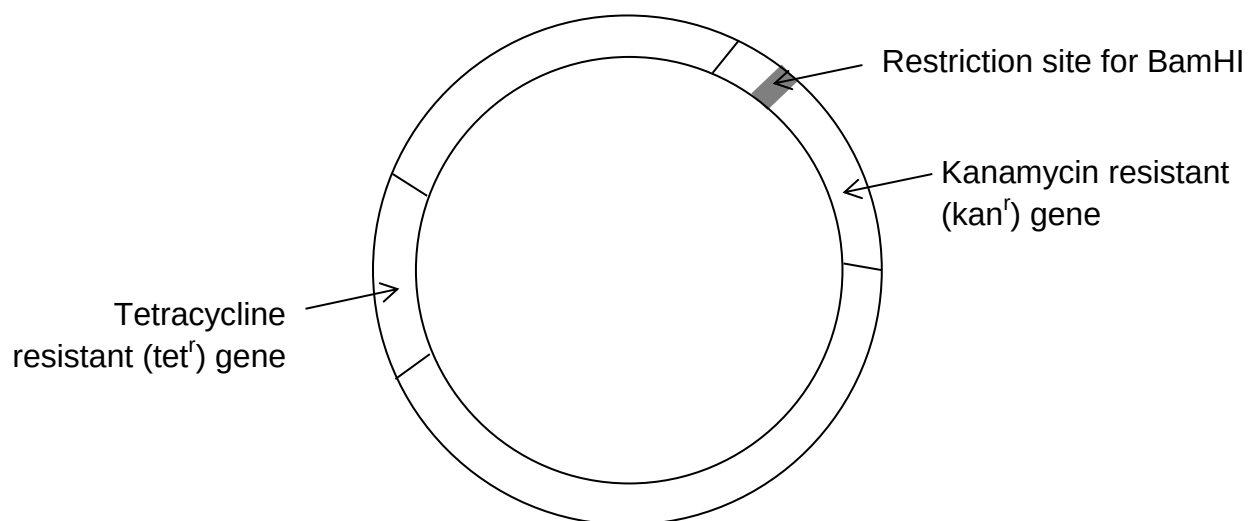


Fig. 4.2

(b) With reference to the properties of bacterial plasmids, provide one reason why this plasmid in Fig 4.2 will not work well as a cloning vector.

.....[1]

1. no origin of DNA replication ;
2. no independent replication of the plasmid (and target gene)
/ cannot result in multiple copies of the plasmid (and target gene) within one bacterium;
OR
3. no prokaryotic promoter;
4. which allows recognition by host cell's RNA polymerase (which is also prokaryotic)
/no transcription / expression of genes;

(c) The student researcher has since made improvements to the plasmid in Fig. 4.2 such that it is now a functional cloning vector.

Steps were then taken to create a recombinant plasmid using BamHI and the improved plasmid.

Restriction enzyme	Specific recognition site
BamHI	5'-G [^] G A T C C-3' 3'-C C T A G [^] G-5'

[^] indicates where the restriction enzyme cuts

(i) Describe the natural function of BamHI restriction enzyme.

.....[1]

- 1 protect bacteria from attack by viruses/bacteriophages ;
- 2 cleave any foreign DNA that enters bacterial cell;/ by cutting up intruding DNA from other organism

(ii) Explain the main problem with using Sample 2 and BamHI in the formation of a recombinant plasmid and suggest how this problem can be solved.

.....[1½]

1. cDNA has blunt ends;
2. ref. incompatibility with sticky ends of plasmid after BamHI cuts;
3. ligate linker DNA;
4. to both ends of cDNA fragment;

(iii) Describe the subsequent steps in the formation of recombinant plasmid.

.....[2]

1. Use the same restriction enzyme to digest **HGH cDNA and plasmid**;
2. to generate complementary sticky ends;
3. The plasmid and the HGH cDNA are mixed together;
4. the sticky ends will anneal by complementary base pairing;
5. DNA ligase added to seal the nicks by catalyzing the formation of phosphodiester bonds (between adjacent nucleotides on the sugar phosphate backbone);

(iv) With reference to Fig. 4.2, explain how you would identify transformed bacterial cells that have taken up the recombinant plasmid with the HGH gene using antibiotic selection.

.....[3]

- 1 grow bacteria on LB/tet plate ;
- 2 select for transformed cells ;
- 3 tetracycline resistance gene intact / survive on LB/tet plate;
- 4 ref. create a replica plate;
- 5 and place on LB/kanamycin plate;
- 6 bacterial colonies that cannot survive on LB/kan plates;
- 7 have the recombinant plasmid;
- 8 insertion of human gene disrupts sequence of kanamycin resistance gene;
- 9 ref. probing for HGH gene;

Escherichia coli is naturally susceptible to ampicillin, which specifically targets cell wall synthesis. The student researcher accidentally spilled ampicillin onto an agar plate containing wild type *E.coli*. Some colonies remained one day after the incident.

(v) Explain the change in allele frequencies in the bacteria population caused by this event.

.....[2½]

- 1 High frequency / over-representation of ampicillin resistant alleles in population /ampicillin susceptible alleles may be lost;
- 2 Bottleneck event;
- 3 Genetic drift occurs; /chance event;
- 4 Small population of survivors which are mainly ampicillin resistant
- 5 ref. these bacteria are at selective advantage
- 6 survive, reproduce and pass down resistant **alleles** to next generation;

Section B

Answer **one** question.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 5 (a) Explain how meiosis and fertilization in sexual reproduction lead to variation within a population. [6]
- (b) Discuss the importance of genetic variation in natural selection and evolution. [4]
- (c) Discuss the difficult ethical concerns of the human genome project. [10]

[Total: 20]

- 6 (a) Describe the molecular structure of starch (amylose) and cellulose and relate these structures to their functions in living organisms. [7]
- (b) Contrast Krebs cycle and Calvin cycle. [6]
- (c) Describe the characteristics of the enzyme active site and the mode of action of enzymes. [7]

[Total: 20]

5 (a) Explain how meiosis and fertilization in sexual reproduction lead to variation within a population. [6]

Crossing over

1. crossing over occurs;
2. between **non-sister chromatids of homologous chromosomes**;
3. during prophase I;
4. where segments of non-sister chromatids exchange positions with each other;
5. ref. chiasmata / synapsis;
6. (results in) new combinations of alleles on the chromosome;

Independent assortment

7. homologous chromosomes/ maternal and paternal chromosome are **randomly aligned on the equator of the spindle**;
8. during metaphase I;
9. ref. independent / random assortment;
10. results in different combinations of chromosomes in daughter cells;
11. $2n$ different combination of gametes formed,
12. n being the haploid number of the organism

Random fusion of gametes

13. random fusion of gametes ;
14. which are genetically different;
15. gives rise to different combinations of alleles in offspring;

Random mating

16. random mating (within population) ;

5(b) Discuss the importance of genetic variation in natural selection and evolution. [4]

1. Genetic variations may be caused by **spontaneous mutations**
2. results in **phenotypic variation**;
3. **creates new alleles** and increases the gene pool for natural selection to operate;
4. (Phenotypic) variation is needed for **natural selection** to act on;
5. Populations with variations are generally better adapted to changes in environment;
6. Different environment have different selection pressures;
7. selects for individuals with phenotype more suited to the existing environment / individuals who are well-adapted to the environment have a **selective advantage** over those who are not;
8. Resulting in the individual able to **survive till reproductive age and reproduce** successfully; passed down **certain alleles to the next generation**;
9. Results in change in allele frequency over time;
10. In the absence of variation, selection pressure acts equally on all individuals;

5(c) Discuss the difficult ethical concerns of the human genome project. [10]

1. **Privacy** of genetic information (who owns, who can control)
2. **Fairness in the use** of genetic information, in non-health care settings, including the potential for misuse.
3. Eg. insurers, employers, courts, schools, adoption agencies, the military (who should have access)
/ e.g. Forensic DNA databanks. Ensuring that they are used for the purpose for which they were collected and protected from misuse. Also, where the public has also assisted the police by volunteering genetic samples to assist in the investigations of unsolved crimes, ensure that special protections are put in place for the DNA samples and the information generated
4. **Patenting of DNA sequences** / commercialization of products
5. may limit their accessibility and development into useful products
6. intellectual property issues surrounding access to and use of genetic information by both researchers and the public)
7. **Fairness in the accessibility** of data/material/technologies
8. may lead to inequity between developed and undeveloped countries to improve human health
9. **Clinical issues** including the education of doctors and other health-service providers, people identified with genetic conditions, and the general public about capabilities, limitations, and social risks and implementation of standards and quality-control measures
10. Ethical issues surrounding the design and conduct of genetic research with people, including the process of **informed consent**.
11. The integration of new genetic technologies, such as **genetic testing**, into the practice of clinical medicine.
12. **Uncertainties** associated with gene tests for susceptibilities and complex conditions (e.g., heart disease) linked to multiple genes and gene-environment interactions.
/ **Accuracy, reliability, and utility of genetic test**
/ Eg. for pre-natal decision making, like abortion
13. Reproductive issues such as the **adequacy of information** for parents/individuals and the rights for decision making for complex and potentially controversial procedures
14. Inappropriate applications of genetic testing such as for the sole purpose of family balancing (sexing of a fetus for this reason)
15. or its use in paternity testing without the informed consent of all parties involved
16. Possible discrimination for individuals with known genetic defects/prone to genetic diseases in different areas of life

/ psychological impact and **stigmatization** due to an individual's genetic differences

17. Use of genetic data to define racial groups, or of racial categories to classify biological traits, is prone to misinterpretation
18. The social implications for both individuals and society, of discovering genetic contributions related to human traits and behaviors such as stigmatized by the suggestion that **alleles associated with what some people perceive as 'negative' physiological or behavioural traits** are more frequent in certain populations
19. Conceptual and philosophical implications regard **human responsibility and degree of genetic influence on behaviour** and hence the party responsible. (Reject “playing God”)
20. The **line between medical disease and enhancement** is controversial. Need for setting of boundaries in applications of the genetics technology.

6(a) Describe the molecular structure of starch (amylose) and cellulose and relate these structures to their functions in living organisms. [7]

Starch

- 1 amylose is unbranched;
- 2 polymer of α -glucose molecules;
- 3 linked by $\alpha(1, 4)$ glycosidic bonds;
- 4 hydrophilic hydroxyl groups of glucose residues project into interior of helices;
- 5 **insoluble**;
- 6 can be stored in large quantities without having any great effect on the water potential of cells
- 7 and can be prevented from diffusing out of cells
- 8 no cross-linking between amylose or glycogen chains
/ no cross-linking to form big bundles
- 9 individual chains can fold into **compact shape**;
- 10 can be stored in large quantities (within a cell);
- 11 **easily hydrolysed** to monosaccharides when required;

Cellulose

- 12 unbranched;
- 13 polymer of β -glucose molecules;
- 14 linked by $\beta(1, 4)$ glycosidic bonds;
- 15 hydroxyl groups project outwards from each cellulose chain;
- 16 form H-bonds with neighboring chains
/ cross-linking of chains to form microfibrils;
- 17 **great tensile strength**
- 18 prevents plant cells from bursting when placed in solutions of higher water potential
/ used as cell wall material for structural support in plants;
- 19 **large intermolecular spaces between microfibrils**
- 20 allows passage of water and solute molecules;

6(b) Contrast Krebs cycle and Calvin cycle.

[6]

Differences

Krebs Cycle	Calvin Cycle
1 occurs in matrix of mitochondria	2 occurs in stroma of chloroplasts
3 catabolic reaction / breakdown of pyruvate	4 anabolic reaction / formation of triose phosphate / glucose / carbohydrates
5 <u>ATP synthesised</u> by substrate level phosphorylation.	6 <u>ATP used</u> in the reduction of PGA to PGAL, and the regeneration of RuBP.
7 Electron carrier = <u>NAD⁺</u> , FAD	8 Electron carrier = <u>NADP⁺</u>
9 Carbon dioxide <u>released</u> by oxidative decarboxylation.	10 Carbon dioxide is <u>fixed</u> by Rubisco.
11 1 cycle => net loss of 2 C	12 1 cycle – net gain of 1 C
13 regenerates oxaloacetate	14 regenerates RuBP

6(c) Describe the characteristics of the enzyme active site and the mode of action of enzymes. [7]

Characteristics of the active site

1. Active site has a specific shape that is complementary to the shape of the substrate;
2. Contact/binding residues: responsible for the specificity of the enzyme;
3. Catalytic residues: residues that act on the bonds in the substrate / responsible for the ability of the enzyme to catalyse a particular reaction;

Mode of action

4. Enzymes work by lowering activation energy required for the reaction;
5. By forming a short-lived enzyme-substrate complex;
They lower activation energy by:
6. Serving as a template to position substrate molecules in the correct orientation for reaction involving two or more substrates to occur;
7. Inducing stress in bonds of substrate;
8. Due to formation of brief bonding e.g. hydrogen bonds and/or ionic bonds with substrate
9. Increasing substrate reactivity;
10. ref.R groups of amino acid residues at the active site change the charge on the substrate/alter the distribution of electrons within the bonds of the substrate/provides appropriate microenvironment;

Lock and Key hypothesis

11. lock-and-key-hypothesis;
12. substrate referred to as key while enzyme is referred to a lock (thus, active sites complementary to substrate);

Induced-fit hypothesis

13. induced-fit hypothesis;
14. initial shape of active site of enzyme might not be complementary to shape of substrate;
15. binding of substrate to enzyme causes an alteration in the conformation of active site;
16. enabling better fit / more effective binding of substrate;
17. ref. successful/effective enzyme-substrate collision
/ ref. correct orientation between active site and substrate;